

Notes: McDonald and Siegel (QJE, 1986)

The Value of Waiting to Invest

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1 Big Picture

1.1 Research question

When should a firm undertake an irreversible investment when the project value and the investment cost are uncertain? The paper's answer is that the standard net-present-value rule, "invest when $V - F \geq 0$," is generally too aggressive. Because investment is irreversible but waiting is reversible, the firm owns an option to invest. The optimal rule is to wait until benefits exceed costs by a strictly positive wedge.

In the paper's notation, V_t is the present value of expected future net cash flows if the project is installed at time t , and F_t is the cost of installation. The firm invests only when the ratio V_t/F_t reaches a critical boundary $C^* > 1$.

1.2 Incremental contribution

This paper is one of the foundational real-options papers. Its incremental contribution is not merely the observation that irreversibility matters; rather, it gives a tractable continuous-time solution for an irreversible investment option when both project value and investment cost are stochastic. It also shows how to value the option under risk aversion when the owners are well diversified, and it uses numerical examples to show that the option value can be large in economically plausible cases.

The paper contributes in five specific ways:

1. It converts the investment timing problem into an option-pricing problem with a first-passage boundary.
2. It derives an explicit formula for the value of the option to invest when V and F follow geometric Brownian motion and the option is infinitely lived.
3. It shows how to compute the relevant discount rate under systematic risk, rather than assuming risk neutrality.
4. It extends the same logic to a scrapping decision and to a case where project value can jump to zero.
5. It demonstrates quantitatively that the hurdle ratio V/F can easily be around 2 or higher, so accepting a project at zero NPV can destroy substantial option value.

1.3 Core intuition

The asymmetry is the entire mechanism. If the firm invests today, it gives up the opportunity to observe future favorable or unfavorable movements in V/F . If it waits, it still has the right to invest later if the state improves, while avoiding losses if the state deteriorates. This option becomes more valuable when uncertainty in V/F is higher, especially when V and F move imperfectly together.

The model therefore changes the meaning of a "profitable" project. A positive static NPV is not sufficient. The project must beat both the cost of investment and the opportunity cost of giving up the option to wait.

2 Model Setup

2.1 Objects and assumptions

The paper studies a project with irreversible installation. At any time t , the firm can pay F_t and receive a project with present value V_t . The baseline model assumes

$$\begin{aligned}\frac{dV}{V} &= \alpha_v dt + \sigma_v dz_v, \\ \frac{dF}{F} &= \alpha_f dt + \sigma_f dz_f,\end{aligned}$$

where dz_v and dz_f are Wiener innovations with instantaneous correlation ρ_{vf} .

The key state variable is not V or F separately but their ratio V/F . The option is homogeneous of degree one in (V, F) , so a proportional scaling of both values does not change the stopping threshold.

2.2 Equation Panel 1: Stochastic Processes for Project Value and Investment Cost / Equation Panel 2: Option Value and Optimal Investment Boundary

II. THE INVESTMENT PROBLEM

We study the investment decision of a firm that is considering the following investment opportunity: at any time t (up to a possible expiration date T), the firm can pay F_t to install an investment project, for which expected future net cash flows conditional on undertaking the project have a present value V_t . We emphasize that V_t is a present value; it represents the appropriately discounted *expected* cash flows, given the information available at time t . For the firm, V_t represents the market value of a claim on the stream of net cash flows that arise from installing the investment project at time t . Typically, V_t and F_t are stochastic. The installation of capacity is irreversible, in that the capacity can be used only for this specific project.

We assume that V_t follows geometric Brownian motion of the form,

$$(1a) \quad \frac{dV}{V} = \alpha_v dt + \sigma_v dz_v,$$

where z_v is a standard Wiener process. Thus, the firm knows the present value of future net cash flows if it installs the project today. However, the present value may be different if the capacity is installed in the future. (We shall also consider the possibility that at some [random] time in the future, the present value of net cash flows drops at once to zero.) Similarly, we assume that F_t follows:

$$(1b) \quad \frac{dF}{F} = \alpha_f dt + \sigma_f dz_f.$$

Equation Panel 1. The paper models project value V_t and investment cost F_t as geometric Brownian motions.

because the optimal rule is to invest when V_t/F_t reaches a barrier C^* , the expected present value of the payoff is

$$E_0[F_{t'} [C^* - 1] e^{-\mu t'}] = [C^* - 1] E_0 [F_{t'} e^{-\mu t'}], \quad (3)$$

where the expectation is taken over the joint density of F_t and the first-passage times for V_t/F_t . Fortunately, it is not necessary to derive the joint density for $F_{t'}$ and t' in order to evaluate (3). The evaluation of (3) is involved, however, so it is relegated to the Appendix. From the Appendix the value of the opportunity is

$$(4) \quad X = (C^* - 1) F_0 \left(\frac{V_0 F_0}{C^*} \right)^\varepsilon,$$

where

$$(5) \quad \varepsilon = \sqrt{\left(\frac{\alpha_v - \alpha_f}{\sigma^2} - \frac{1}{2} \right)^2 + \frac{2(\mu - \alpha_f)}{\sigma^2}} + \left(\frac{1}{2} - \frac{\alpha_v - \alpha_f}{\sigma^2} \right),$$

$$C^* = \varepsilon / (\varepsilon - 1),$$

$\sigma^2 = \sigma_v^2 + \sigma_f^2 - 2\rho_{vf}\sigma_v\sigma_f$, and ρ_{vf} is the instantaneous correlation between the rates of increase of V and F .

Equation Panel 2. The option value formula and stopping threshold C^ for the infinitely lived investment opportunity.*

Interpretation. Equation Panel 1 states the uncertainty environment. The project is not evaluated once and for all; both the value of the installed project and the installation cost evolve

continuously. The geometric Brownian motion structure is what gives the paper a closed-form solution, but it also later creates some limitations, because an unbounded GBM may imply that the firm should wait forever in some parameter regions.

Equation Panel 2 gives the paper's central formula. The investment opportunity has value

$$X = (C^* - 1)F_0 \left(\frac{V_0/F_0}{C^*} \right)^\epsilon,$$

with

$$C^* = \frac{\epsilon}{\epsilon - 1}.$$

Because $\epsilon > 1$, the threshold C^* is greater than 1. Thus, the firm waits until V/F exceeds a hurdle ratio, not merely until $V = F$. The option value is increasing in the payoff at exercise, $(C^* - 1)F_0$, and in the probability-adjusted distance of the current ratio V_0/F_0 from the boundary C^* .

2.3 Why the stopping rule depends on V/F

The stopping boundary is a ratio because the problem is scale invariant. If both project value and investment cost double, the decision problem is identical except for scale. Therefore, the optimal policy cannot depend on levels alone. It must depend on V/F .

This feature is important for empirical interpretation. A project with a large dollar NPV can still be premature if both V and F are large but V/F has not crossed the required threshold. Conversely, a small project can be optimal if its benefit-to-cost ratio is high enough.

3 Risk Adjustment and Discounting

3.1 Risk-adjusted return logic

The paper does not simply discount the future payoff at an arbitrary rate. Instead, it assumes the owners of the option are well diversified, so only systematic risk is priced. The option's unanticipated return is a weighted average of the unanticipated return components in V and F . This allows the paper to map the option's discount rate to required returns on assets with the same systematic risks as V and F .

Let $\hat{\alpha}_v$ and $\hat{\alpha}_f$ denote the required expected returns on assets with the same risk as V and F . Define

$$\delta_v = \hat{\alpha}_v - \alpha_v, \quad \delta_f = \hat{\alpha}_f - \alpha_f.$$

These δ terms are the shortfalls of expected capital gains relative to required returns. Economically, they are analogous to payout rates or opportunity costs of waiting.

3.2 Equation Panel 3: Risk-Adjusted Discount Rate / Equation Panel 4: Jump-to-Zero Process for Project Value

standard asset pricing models, the risk premium earned on an asset is proportional to the riskiness of the asset. For example, in the Capital Asset Pricing Model,

$$(9) \quad \hat{\alpha}_i - r = \phi \rho_{im} \sigma_i,$$

where $\hat{\alpha}_i$ is the required rate of return on asset i , r is the risk-free rate, ϕ is the market price of risk, and ρ_{im} is the correlation between the rate of return on the asset and that on the market portfolio. If $\hat{\alpha}_j$ is the required rate of return for an asset with unexpected rate of return $\sigma_j dz_j$, then from (9) an asset with unexpected rate of return $\nu \sigma_j dz_j$ will have required rate of return $r + \nu(\hat{\alpha}_j - r)$. Therefore, μ , which is the discount rate for future payoffs and hence the equilibrium expected rate of return on the investment opportunity, will be given by

$$(10) \quad \mu = r + \varepsilon(\hat{\alpha}_v - r) + (1 - \varepsilon)(\hat{\alpha}_f - r) = \varepsilon \hat{\alpha}_v + (1 - \varepsilon) \hat{\alpha}_f,$$

where $\hat{\alpha}_v$ and $\hat{\alpha}_f$ are determined by (9), and are the expected rates of return required by investors for assets that are perfectly correlated with, and have the same standard deviations as V and F .

Equating the required expected rate of return (10) with the actual expected rate of return on X in (8) yields a quadratic equation in ε :

$$(11) \quad \mu = \varepsilon \hat{\alpha}_v + (1 - \varepsilon) \hat{\alpha}_f = \varepsilon \alpha_v + (1 - \varepsilon) \alpha_f + \frac{1}{2} \varepsilon (\varepsilon - 1) \sigma^2.$$

Equation (11) has the solution,

$$(12) \quad \varepsilon = \sqrt{\left(\frac{\delta_f - \delta_v}{\sigma^2} - \frac{1}{2}\right)^2 + 2 \frac{\delta_f}{\sigma^2} + \left(\frac{1}{2} - \frac{\delta_f - \delta_v}{\sigma^2}\right)},$$

where $\delta_v = \hat{\alpha}_v - \alpha_v$ and $\delta_f = \hat{\alpha}_f - \alpha_f$. $\delta_v > 0$ insures that $\varepsilon > 1$.

Equation Panel 3. The option's required return is constructed from the systematic risks of V and F .

Interpretation. Equation Panel 3 shows how the model avoids a purely risk-neutral treatment. The key object is the required return on the option to invest, which is a weighted average of the required returns on the underlying stochastic variables. The resulting formula for ε depends on δ_v , δ_f , and the variance of V/F . Thus, the drift terms α_v and α_f matter only relative to the required returns on assets with similar risks.

Equation Panel 4 extends the model by allowing the project value to experience a Poisson jump to zero. This is economically useful for projects whose opportunity can disappear suddenly, such as a technology that becomes obsolete, a product that is preempted, or a market opportunity that vanishes. The jump risk makes waiting less attractive because the option itself can expire unexpectedly.

Jumps in V_t

Assume that there is a positive probability that the present value of net future cash flows, V_t , can take a discrete jump to zero. If this happens, the investment opportunity becomes worthless. Thus the stochastic process for V_t is a mixed Poisson-Wiener process of the form,

$$(15) \quad \frac{dV}{V} = \alpha_v dt + \sigma_v dz_v + dq,$$

where

$$dq = \begin{cases} -1 & \text{with probability } \lambda dt \\ 0 & \text{with probability } 1 - \lambda dt. \end{cases}$$

The occurrence of the Poisson event induces the process to stop, since zero is a natural absorbing barrier for a geometric Brownian motion process.

Notice that when the Poisson event occurs, it is as if the investment opportunity expires, since its value becomes zero. Thus, calculating the value of the investment opportunity when V_t can jump to zero is like calculating the value of an investment opportunity with an uncertain expiration date. The value in this case is easily calculated thanks to a result in Merton [1971].

The distribution of first occurrence times for a Poisson event with parameter λ is exponential. Suppose that the Poisson event is uncorrelated with the market portfolio and V , and that F is constant. Then the expected present value of the payoff from the investment opportunity with uncertain expiration date is

$$(16) \quad X^* = \int_0^\infty \lambda e^{-\lambda T} X(T) dT.$$

This may be integrated by parts [Merton, 1971] to give

$$(17) \quad X^* = \max E_0(e^{-(\mu + \lambda)T} F | C; -1).$$

Equation Panel 4. The extension in which project value can jump discretely to zero.

3.3 Equation Panel 5: Jump Risk and Discount Adjustment / Equation Panel 6: Monopolist Example and Endogenous Project Value

by $\mu + \lambda$.¹⁰ The formula is therefore the same as (4) with the discount rate adjusted in (5).

To compute the discount rate in this case, we can use Ito's lemma for Poisson processes [Merton, 1971] to calculate the expected rate of return on the option to invest. Assume for simplicity that F is fixed. The Ito derivative of (4) is then

$$(18) \quad \frac{dX^*}{X^*} = \varepsilon(\alpha_v dt + \sigma_v dz_v) + \varepsilon(\varepsilon - 1) \frac{1}{2} \sigma_v^2 dt - \lambda dt.$$

The only risk in (18) is due to the term $\varepsilon \sigma_v dz_v$, so the required rate of return on X^* is $r + \varepsilon(\hat{\alpha}_v - r)$. Equating this to the expected rate of return gives

$$(19) \quad [r + \varepsilon(\hat{\alpha}_v - r)]dt = E\left(\frac{dX^*}{X^*}\right) = [\varepsilon\alpha_v - \lambda + \varepsilon(\varepsilon - 1) \frac{1}{2} \sigma_v^2]dt.$$

Solving for ε yields

$$(20) \quad \varepsilon = \sqrt{\left(\frac{r - \delta_v}{\sigma_v^2} - \frac{1}{2}\right)^2 + \frac{2(r + \lambda)}{\sigma_v^2}} + \left(\frac{1}{2} - \frac{r - \delta_v}{\sigma_v^2}\right).$$

For the case when F is random, every occurrence of r in (20) is replaced by δ_f .

Equation Panel 5. Jump risk enters as an additional discounting force and lowers the value of waiting.

where $\bar{K}$ is the fixed level of capital, and Q_t and L_t are quantity produced and labor employed at time t . The firm faces an inverse demand curve given by

$$(22) \quad P_t = \theta_t Q_t^{-1/\eta},$$

where P_t is the price of the commodity at time t , η is the price elasticity of demand, and θ_t is a demand shift parameter following the stochastic process,

$$(23) \quad \frac{d\theta}{\theta} = \alpha_\theta dt + \sigma_\theta dz_\theta.$$

Let $\rho_{\theta m}$ be the correlation of θ with the market portfolio.

After the project is installed, instantaneous profits are given by $\pi_t = P_t Q_t - \bar{w} L_t$, where $\bar{w}$ is the wage. At each t , the firm chooses labor usage to maximize instantaneous profits. Maximized profits at t are then $B\theta_t^\gamma$, where $\gamma = [1 - \beta(1 - 1/\eta)]^{-1} > 1$ and B is a constant. When the project lives forever, the present value of expected maximized profits is

$$(24) \quad V(\theta_0) = B\theta_0^\gamma / (\hat{\alpha}_v - \alpha_v),$$

where $\alpha_v = \gamma\alpha_\theta + \frac{1}{2}\gamma(\gamma - 1)\sigma_\theta^2$ and

$$(25) \quad \hat{\alpha}_v = r + \phi\rho_{\theta m}\gamma\sigma_\theta,$$

which follows from (9) together with the fact that $\sigma_v = \gamma\sigma_\theta$ and $\rho_{\theta m} = \rho_{vm}$. Equation (24) can be rewritten as

$$(26) \quad \hat{\alpha}_v = \alpha_v + B\theta_0^\gamma / V_0 = \alpha_v + \delta_v,$$

so that δ_v represents the payout ratio of the installed project.

Table IV displays the option values and investment rules corresponding to different values of the underlying exogenous parameters. It is important to realize that the comparative static experiment in this table is different from that in Tables I and II.

Equation Panel 6. A monopolist example derives V from production, demand, and systematic demand risk.

Interpretation. Equation Panel 5 makes the jump-to-zero extension operational. The jump arrival rate λ behaves like an increase in the effective discount rate. The higher the probability that the opportunity disappears, the less valuable it is to wait, and the lower the investment threshold C^* becomes.

Equation Panel 6 links the abstract project value V to an economic environment. A monopolist faces stochastic demand and chooses labor after installation. The present value of profits becomes a function of the demand shifter θ . This example is useful because it shows that the reduced-form variables in the main formula can be derived from primitives such as demand elasticity, technology, volatility, and correlation with the market portfolio.

4 Main Numerical Results

4.1 Table I: Value of Investment Opportunity When $V = F = 1$ / Table II: Value of Benefits Relative to Investment Cost (V/F) at Which Investment Is Optimal

TABLE I
VALUE OF INVESTMENT OPPORTUNITY WHEN $V = F = 1$

δ_v	0.05			0.10			0.25		
ρ_{vf}	-0.5	0.0	0.5	-0.5	0.0	0.5	-0.5	0.0	0.5
σ_v^2, σ_f^2									
0.01	0.33	0.30	0.28	0.14	0.12	0.08	0.03	0.02	0.01
0.02	0.38	0.34	0.30	0.20	0.16	0.12	0.06	0.04	0.02
0.04	0.45	0.40	0.34	0.27	0.23	0.16	0.11	0.08	0.04
0.10	0.57	0.51	0.43	0.40	0.34	0.25	0.21	0.16	0.09
0.20	0.67	0.61	0.51	0.52	0.45	0.34	0.32	0.25	0.16
0.30	0.73	0.67	0.57	0.60	0.52	0.40	0.39	0.32	0.21
λ									
0.00	0.45	0.40	0.34	0.27	0.23	0.16	0.11	0.08	0.04
0.05	0.33	0.29	0.24	0.23	0.19	0.13	0.10	0.07	0.04
0.10	0.27	0.23	0.19	0.20	0.16	0.12	0.10	0.07	0.04
0.25	0.19	0.16	0.12	0.15	0.12	0.09	0.09	0.06	0.04
δ_f									
0.01	0.30	0.23	0.14	0.18	0.13	0.07	0.08	0.06	0.03
0.05	0.37	0.31	0.23	0.22	0.17	0.10	0.09	0.06	0.03
0.10	0.45	0.40	0.34	0.27	0.23	0.16	0.11	0.08	0.04
0.25	0.60	0.58	0.56	0.42	0.39	0.36	0.18	0.15	0.10

Note. Entries are calculated using (4) and (12) in the text. Base case parameters are $\sigma_v^2 = \sigma_f^2 = 0.04$; $\delta_v = \delta_f = 0.10$; $\lambda = 0.00$.

Table I. Value of investment opportunity when current benefits equal current investment cost.

TABLE II
VALUE OF BENEFITS RELATIVE TO INVESTMENT COST (V/F)
AT WHICH INVESTMENT IS OPTIMAL

δ_v	0.05			0.10			0.25		
ρ_{vf}	-0.5	0.0	0.5	-0.5	0.0	0.5	-0.5	0.0	0.5
σ_v^2, σ_f^2									
0.01	2.50	2.35	2.18	1.47	1.37	1.25	1.09	1.06	1.03
0.02	2.91	2.64	2.35	1.72	1.56	1.37	1.18	1.12	1.06
0.04	3.65	3.17	2.64	2.13	1.86	1.56	1.34	1.24	1.12
0.10	5.65	4.56	3.41	3.19	2.62	2.00	1.77	1.54	1.29
0.20	8.77	6.70	4.56	4.79	3.73	2.62	2.44	2.00	1.54
0.30	11.83	8.77	5.65	6.34	4.79	3.19	3.07	2.44	1.77
λ									
0.00	3.65	3.17	2.64	2.13	1.86	1.56	1.34	1.24	1.12
0.05	2.50	2.23	1.92	1.86	1.67	1.44	1.32	1.23	1.12
0.10	2.10	1.90	1.67	1.72	1.56	1.37	1.30	1.22	1.12
0.25	1.67	1.54	1.40	1.51	1.40	1.27	1.27	1.19	1.11
δ_f									
0.01	2.31	1.89	1.46	1.64	1.43	1.22	1.25	1.17	1.08
0.05	2.85	2.38	1.86	1.83	1.58	1.32	1.28	1.19	1.10
0.10	3.65	3.17	2.64	2.13	1.86	1.56	1.34	1.24	1.12
0.25	6.42	5.96	5.49	3.35	3.09	2.81	1.62	1.49	1.33

Note. Entries are calculated using equations (4) and (12) in the text. Base case parameters are $\delta_v = \delta_f = 0.10$; $\sigma_v^2 = \sigma_f^2 = 0.04$; $\lambda = 0.00$.

Table II. Critical V/F ratio at which investment becomes optimal.

Table I interpretation. Table I reports the value of the option to invest when the static NPV is zero, $V = F = 1$. Since immediate investment at $V = F$ would deliver zero payoff, the table measures the value destroyed by exercising too early. The magnitudes are economically large. In the benchmark neighborhood, the option can be worth around 0.16 to 0.23 per dollar of project value; under higher uncertainty, it can be much larger.

The comparative statics are central:

- Higher volatility in V/F raises option value because waiting gives upside exposure while limiting downside exposure.
- Lower correlation between V and F raises option value because the ratio V/F becomes more volatile.
- Higher δ_v tends to reduce the option value because waiting forgoes more current project payoff.
- A higher jump hazard λ lowers the option value because the opportunity may disappear before it is exercised.
- Higher δ_f can raise or alter the option value because the cost side of the option changes relative to the project-value side.

Table II interpretation. Table II converts the option value into an investment hurdle. The critical ratio V/F is often well above 1. In the example emphasized in the text, with $\sigma_v^2 = \sigma_f^2 = 0.02$, $\rho_{vf} = 0$, and $\delta_v = \delta_f = 0.10$, investment is optimal only when V/F reaches about 1.56. Under more volatile parameter values, the threshold can be far higher. The table therefore provides the paper's practical message: a project can have positive static NPV and still be worth delaying.

4.2 Table III: Percentage of Value of Investment Option Which Is Due to Uncertainty / Table IV: Option Values and Investment Rules

TABLE III
PERCENTAGE OF VALUE OF INVESTMENT OPTION WHICH IS DUE TO UNCERTAINTY

	$\sigma^2 = 0.02$	$\sigma^2 = 0.04$	$\sigma^2 = 0.10$	$\sigma^2 = 0.30$
$\delta_v = 0.02$	5.3	9.4	17.2	28.4
$\delta_v = 0.04$	12.5	20.1	32.6	47.1
$\delta_v = 0.06$	25.6	36.4	50.8	64.6
$\delta_v = 0.08$	51.4	61.9	73.0	82.1
$\delta_v = 0.10$	100.0	100.0	100.0	100.0

Note. Entries are calculated using equations (4), (12), and (13) in the text. All of these calculations assume that $V = F = 1$ and $\delta_f = 0.10$.

Table III. Share of the option value attributable to uncertainty rather than deterministic waiting.

TABLE IV
OPTION VALUES AND INVESTMENT RULES

α_0		-0.03		0.00				0.01	
ρ_{vm}	0.00	0.20	0.50	0.00	0.20	0.50	0.00	0.20	0.50
σ_v^2									
0.01	0.0740 (1.22)	0.0597 (1.18)	0.0459 (1.13)	0.1890 (1.68)	0.1305 (1.43)	0.0837 (1.26)	0.2902 (2.25)	0.1890 (1.68)	0.1110 (1.35)
0.04	0.2227 (1.85)	0.1657 (1.58)	0.1177 (1.38)	0.4127 (3.28)	0.2659 (2.09)	0.1657 (1.58)	0.5493 (5.28)	0.3257 (2.50)	0.1905 (1.69)
0.10	0.4399 (3.58)	0.3044 (2.35)	0.2056 (1.76)	0.7545 (13.7)	0.4302 (3.47)	0.2592 (2.05)	*	0.4973 (4.36)	0.2835 (2.20)
0.20	0.7647 (14.6)	0.4736 (4.02)	0.3053 (2.35)	*	0.6318 (7.37)	0.3631 (2.80)	*	0.7151 (11.0)	0.3874 (3.02)

Note. Values are option values computed using (4) and (12) in the text. Values in parentheses are optimal investment levels. Technological parameters are chosen such that $V = 1$. F is nonstochastic and equal to 1. Other parameter values are $\eta = 2.0$; $r = 0.05$; $\beta = 2.3$; and $\phi = 0.50$.

*For these values $\delta_v < 0$, so the option value is undefined.

Table IV. Option values and investment thresholds in the monopolist example.

Table III interpretation. Table III separates the value of waiting under certainty from the extra value generated by uncertainty. When uncertainty is small and the opportunity cost of waiting is low, some value can arise from deterministic timing. But as uncertainty rises, a larger share of the investment option's value comes from the ability to capture upside states while avoiding downside states. The 100 percent entries show cases where waiting would have no value without uncertainty.

Table IV interpretation. Table IV embeds the real-option logic in the monopolist example. The entries outside parentheses are option values; the values in parentheses are critical investment thresholds. Higher demand volatility generally increases both the option value and the waiting threshold. Higher correlation with the market portfolio can reduce option value because it raises the systematic risk component of project value and therefore the required return. The starred entries occur when the GBM specification implies an undefined option value or effectively infinite waiting; this is not treated as a realistic prediction but as a limitation of the GBM assumption.

5 Economic Mechanism

5.1 Why uncertainty raises the investment hurdle

The option is convex in the ratio V/F . If V/F rises, the firm can invest and lock in the gain. If V/F falls, the firm simply continues to wait. This asymmetry gives volatility a positive value. The value of the option increases with the variance of proportional changes in V/F ,

$$\sigma^2 = \sigma_v^2 + \sigma_f^2 - 2\rho_{vf}\sigma_v\sigma_f.$$

A high correlation between V and F dampens changes in V/F , reducing the value of waiting. A low or negative correlation makes the ratio more volatile, increasing the value of waiting.

5.2 Why the NPV rule fails

The NPV rule compares investing now to never investing. The correct real-options comparison is investing now versus retaining the right to invest later. The latter alternative is valuable whenever uncertainty remains and the decision is irreversible.

Therefore, the correct decision is not

$$\text{Invest if } V - F \geq 0,$$

but rather

$$\text{Invest if } \frac{V}{F} \geq C^*.$$

The wedge $C^* - 1$ is the value of the option being sacrificed by exercise.

5.3 Scrapping as the mirror problem

The same mathematics applies to scrapping. If V is the value of keeping the project and F is the scrap value, the firm should not scrap as soon as $F > V$. It should wait until the scrap value exceeds the continuation value by a sufficient margin. This is the abandonment-option counterpart to the investment-option problem.

6 Key Takeaways

1. **Irreversibility creates an option value of waiting.** The firm should delay investment unless V/F exceeds a critical threshold.
2. **Positive NPV is not enough.** The project must compensate the firm for giving up the option to invest later.
3. **Uncertainty raises the hurdle.** More volatility in V/F makes waiting more valuable.
4. **Systematic risk matters through required returns.** The option is valued as a claim held by diversified investors, so only priced risk affects the discounting formula.
5. **Option values are quantitatively large.** The paper's tables show that premature investment can cost 10–20 percent or more of project value.
6. **The framework is broad.** It applies to investment, scrapping, monopoly investment, and sudden obsolescence risks.

7 Limitations and Caveats

The paper is explicit about several limitations.

First, the infinite-lived option assumption rules out known expiration dates, such as patents or regulatory deadlines. Finite-horizon versions generally require numerical methods.

Second, geometric Brownian motion can be too strong for real quantities such as demand shocks. It allows values to drift without bound and can generate implausible “never invest” cases. Mean reversion would often be more realistic but less analytically tractable.

Third, the investment is assumed to be lumpy and irreversible. If investment can be scaled continuously, made gradually, or partially reversed, the value of waiting may be lower.

Fourth, the model treats the post-investment project value V as already incorporating the value of future operating decisions. In many applications, operating flexibility, abandonment, expansion, and contraction options would need to be modeled jointly.

8 How to Read the Paper

A useful way to read the paper is to keep three levels separate:

1. **Mathematical level:** the investment problem is an exchange option with stochastic V and F .
2. **Economic level:** irreversibility plus uncertainty means that the firm needs a hurdle above zero NPV.
3. **Quantitative level:** the wedge can be large enough to change investment policy in practice.

The main formula is not just a technical result. It is the reason why firms may rationally appear to “delay profitable projects.” Delay is not necessarily managerial inertia; it can be the optimal exercise policy for a valuable real option.

9 Discussion Questions

1. In empirical investment studies, how would ignoring the option value of waiting bias estimates of investment sensitivity to Tobin’s q ?
2. Which real-world projects are closest to the paper’s assumptions: natural resource extraction, patent-protected products, real estate development, or technology adoption?
3. How would the threshold change if the firm faced a known expiration date on the investment opportunity?
4. How would partial reversibility or resale value change the investment boundary?
5. Is geometric Brownian motion a reasonable model for project value, investment cost, and demand shocks in modern applications?

10 One-Sentence Summary

McDonald and Siegel show that an irreversible investment should be treated as a real option, so the firm should wait until project value exceeds investment cost by enough to compensate for the lost value of future flexibility.